QUALITY ASSURANCE PLAN

AuralAid: Adaptive AI Hearing with Contextual Understanding

Prepared as	Quality Assurance Plan tailored to the final year project proposal
Project Domain	AI-enabled assistive hearing aid system with embedded processing, mobile app, captioning, emotion detection, alerts, and cloud/mobile connectivity
Reference Basis	Proposal scope and architecture plus the structure/style used in the supplied QA template
Document Purpose	Define test strategy, quality metrics, traceability, compliance targets, environments, and planned test suites before full validation execution
Document Version	Version 1.0

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1. Quality Assurance Plan

This Quality Assurance Plan defines how AuralAid will be verified and validated as an assistive, AI-enabled hearing aid prototype. The plan is customized to the submitted proposal, which describes an embedded audio-processing device built around the ESP32-S3, mobile or web connectivity, live speech captions, speaker emotion detection, hazardous-noise alerts, and in later technical sections, optional health-monitoring and caregiver alert capabilities. The purpose of this plan is to ensure the delivered prototype is functional, safe to evaluate, reliable under realistic use conditions, and traceable to the project objectives and business success criteria.

Because the project combines hardware, firmware, AI inference, mobile software, communication links, and user-facing assistive behavior, QA will be performed at multiple levels: component testing, integration testing, system testing, usability validation, field-condition testing, and non-functional evaluation. This document keeps the sectioning style of the reference QA template while replacing its web-platform-focused tests with hearing-aid-specific suites, metrics, and acceptance thresholds.

1.1 QA Objectives

- Verify that the prototype captures and processes audio correctly in low-noise and noisy environments.
- Validate that real-time captions are accurate enough for demonstrator use and remain responsive under realistic speech conditions.
- Confirm that speaker-emotion detection is directionally useful, clearly bounded, and does not create misleading or unsafe outputs.
- Verify that noise-alert functionality detects hazardous sound levels and notifies the user within acceptable response time.
- Validate BLE/Wi-Fi/mobile-app interoperability, history display, alert handling, and graceful behavior during disconnections.
- Check that security, privacy, and data-handling controls are appropriate for prototype use, especially for voice, captions, and health-related data.
- Ensure the system remains stable, recoverable, and power-efficient during repeated sessions and extended operation.

1.2 Scope of QA

In scope:

- Embedded hardware and firmware: microphones, DSP/noise filtering pipeline, local alerting, sensor integration, power behavior, and fault recovery.
- AI-enabled features: speech-to-text pipeline, emotion classification, threshold-based noise alerts, summarization or context display if implemented.
- Application layer: mobile app or web dashboard, live caption display, emotion visualization, alert presentation, settings, history, and cloud sync if enabled.
- System qualities: usability, accessibility, latency, robustness, privacy, data integrity, and defect tracking.
- Out of scope for this prototype QA cycle:
- Formal medical-device certification or regulatory submission.
- Clinical efficacy claims, diagnostic claims, or guaranteed therapeutic outcomes.
- Mass-production validation such as full reliability qualification, EMC certification, or formal hearing-aid compliance lab testing.

1.3 QA Approach and Test Levels

Test Level	Purpose
Component Testing	Microphones, ESP32-S3 firmware modules, alert actuators, sensors, BLE services, API endpoints, and app screens are tested individually before integration.
Integration Testing	Audio capture, DSP, STT, emotion inference, app connectivity, cloud/logging, and alert flows are tested end-to-end across hardware and software boundaries.
System Testing	Complete user scenarios are exercised under controlled laboratory conditions and realistic ambient-noise conditions.
Non-Functional Testing	Latency, accuracy, battery usage, stability, security, accessibility, and recovery behavior are validated against thresholds.
User Validation	Representative usability sessions assess readability of captions, clarity of alerts, wearable comfort, and ease of interaction.

1.4 Entry and Exit Criteria

Entry criteria for formal QA execution:

- Baseline firmware and mobile/web application build are available and version-controlled.
- Core hardware bring-up is complete and testable on at least one stable prototype unit.
- Known critical blockers affecting power-up, audio capture, or communication are resolved or explicitly waived.
- Test data, speech scripts, noise samples, and reference devices are prepared.
- Exit criteria for QA sign-off:
- All critical and high-severity defects affecting core assistive functions are closed or formally accepted with workaround.
- Target quality metrics for latency, caption accuracy, noise-alert response, and app stability are met or justified.
- Regression testing passes on the release candidate build.
- Test summary report, defect log, and residual-risk note are reviewed by the supervisor and project team.

1.5 Roles and Responsibilities

Role	Responsibility
QA Lead / Student Tester	Owns this plan, prepares test cases, executes tests, maintains evidence, and reports defects.
Firmware Engineer	Supports hardware/firmware debug, instrumentation, logging, and root-cause analysis.
AI/ML Engineer	Validates dataset usage, threshold tuning, caption and emotion metrics, and model behavior review.
Mobile/Web Developer	Resolves UI, connectivity, history, and API issues; supports regression on the application side.
Project Supervisor	Reviews coverage, approves acceptance thresholds, and signs off major QA milestones.

1.6 Quality Metrics and Acceptance Targets

Metric	Measurement Method	Target / Acceptance Threshold
Caption accuracy	Word-level transcription accuracy/WER on scripted test phrases in quiet and noisy conditions	Target >= 85% accuracy in quiet; >= 75% in controlled noisy scenes for prototype demonstrations
Caption latency	Speech end to caption display	Target <= 2.5 s typical; no critical freezes
Noise filtering benefit	Relative improvement in intelligibility/SNR or listener rating versus raw audio	Target measurable improvement over unfiltered baseline
Emotion detection usefulness	Top-1 or grouped-class accuracy on supported labels	Target >= 70% on supported classes; unclear cases should be labeled neutral/uncertain rather than misleading
Noise-alert response	Time from threshold exceedance to user alert	Target <= 2 s for hazardous-noise alert
Noise-alert correctness	True detection for sounds above configured threshold; false-alert frequency	Target >= 95% detection in lab threshold tests; false alerts controlled through hysteresis
Connectivity reliability	Successful BLE/Wi-Fi pairing, reconnection, and data delivery	Target >= 95% successful connection attempts in controlled tests
Application stability	Crash-free sessions during guided test runs	Target >= 95% crash-free sessions
Battery/runtime	Continuous operation under representative use	Target aligned to project demonstration needs; no unsafe thermal or brownout behavior
User satisfaction	Post-test questionnaire rating on readability/usability	Target >= 80% positive feedback in pilot evaluation

1.7 Compliance and Good-Practice References

This is an academic prototype, not a certified medical device. Even so, QA activities will be aligned with relevant engineering and software good practices so the project remains structured, safe, and credible.

Reference Area	Planned QA Interpretation
Software lifecycle discipline	Development and testing evidence should be organized in a manner inspired by IEC 62304-style traceability for safety-relevant software.
Risk management	Hazard identification, defect triage, and residual-risk discussion should follow an ISO 14971-style risk-management mindset.
Electrical and EMC awareness	Hardware test planning should consider the intent of IEC 60601-1 and IEC 60601-1-2 principles where applicable, without claiming certification.
Usability and accessibility	Mobile/web interface checks should consider accessibility, readability, contrast, and understandable interaction flows consistent with WCAG and general human-factors practice.
Privacy and security	Voice, caption, and health-related data handling should follow least-privilege, encrypted transmission, secure storage, and consent-aware prototype practices.

2. Planned Test Suites and Test Cases

The following test suites are tailored to the capabilities and risks described in the proposal. Unlike the reference template, these suites are organized around an assistive hearing-aid product that blends embedded processing, AI inference, connectivity, and user alerts. All statuses are set to Planned because this document defines the QA baseline prior to full execution.

2.1. Test Suite 01: Device Bring-Up and Hardware Integration

This suite validates the basic integrity of the embedded platform before feature-level testing begins.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Power-On and Boot	Verify that the device powers on safely and reaches ready state.	1. Apply nominal power. 2. Observe boot sequence. 3. Confirm ready indicator/app connection state.	Device boots without reset loops, unsafe heating, or missing services.	Planned
TC-2	Microphone Enumeration	Verify that all required microphones/sensors are detected by firmware.	1. Start firmware diagnostics. 2. Read microphone/sensor status.	All required audio inputs and configured sensors report healthy status.	Planned
TC-3	Audio Path Sanity	Validate raw audio capture before DSP is applied.	1. Record reference speech. 2. Inspect waveform/stream in debug mode.	Audio frames are present, not clipped, and within expected amplitude range.	Planned
TC-4	Actuator Check	Verify vibration/LED/audio alert hardware activation.	1. Trigger each alert mode from diagnostics. 2. Observe output.	Each alert actuator responds correctly and repeatably.	Planned
TC-5	Safe Failure	Validate behavior under sensor disconnect or startup fault.	1. Disconnect one input/sensor. 2. Reboot or run self-test.	System logs the fault, stays recoverable, and does not enter unsafe state.	Planned

2.2. Test Suite 02: Audio Capture and Noise Filtering

This suite focuses on the signal-processing core of the assistive device and its performance in realistic acoustic conditions.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Quiet-Room Capture	Verify clean voice capture in low-noise conditions.	1. Play or speak standard script. 2. Capture processed output.	Speech is clear with no unexpected distortion.	Planned
TC-2	Noise Reduction in Crowd Noise	Assess DSP improvement under cafeteria/street noise.	1. Play reference speech plus controlled noise. 2. Compare filtered vs raw output.	Filtered output is measurably or perceptually better than raw audio.	Planned
TC-3	Non-Speech Suppression	Check that background hum/fan noise is reduced.	1. Feed continuous background noise. 2. Observe processed stream.	Steady background noise is attenuated without audio instability.	Planned

TC-ID	Title	Description	Steps	Expected Result	Status
TC-4	Speech Preservation	Ensure DSP does not overly suppress low-volume speech.	1. Play soft speech samples. 2. Run pipeline.	Important speech remains intelligible after filtering.	Planned
TC-5	Overload Handling	Verify handling of very loud input without crash.	1. Inject high-level sound. 2. Monitor clipping and recovery.	System limits distortion as much as possible and recovers cleanly.	Planned

2.3. Test Suite 03: Speech-to-Text Captioning

This suite validates the real-time captioning feature that is central to communication support for hearing-impaired users.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Caption Generation in Quiet	Validate live caption generation for standard speech.	1. Connect app. 2. Speak scripted phrases. 3. Observe captions.	Captions appear with acceptable delay and good accuracy.	Planned
TC-2	Captioning in Noise	Assess caption robustness in controlled background noise.	1. Repeat scripted phrases with noise playback. 2. Record outputs.	Caption quality degrades gracefully and remains usable.	Planned
TC-3	Accent and Speed Variation	Check behavior for different speaking styles.	1. Use multiple speakers and speaking rates. 2. Compare transcripts.	System handles expected variation without severe failure.	Planned
TC-4	Long Utterance Handling	Validate multi-sentence caption flow.	1. Speak longer passage. 2. Observe segmentation and scroll behavior.	Captions remain readable and do not freeze or truncate unexpectedly.	Planned
TC-5	Recognition Failure Handling	Verify fallback messaging when STT confidence is low.	1. Use unclear speech/noisy audio. 2. Observe UI/logic.	System marks result as uncertain or requests retry instead of misleading output.	Planned

2.4. Test Suite 04: Emotion Detection

This suite verifies that emotion cues are useful, bounded, and presented responsibly as contextual assistance rather than definitive judgment.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Basic Emotion Classes	Validate supported emotion labels on reference samples.	1. Feed curated clips for happy, sad, angry, neutral. 2. Observe outputs.	Predicted label matches supported class or returns neutral/uncertain when ambiguous.	Planned
TC-2	Confidence Thresholding	Check that low-confidence predictions are bounded.	1. Feed ambiguous clips. 2. Review confidence logic.	System avoids overconfident false emotion labels.	Planned
TC-3	Display Consistency	Validate app presentation of emotion feedback.	1. Trigger different classifications. 2. Inspect labels/colors/icons.	Emotion visualization is understandable and consistent across screens.	Planned

TC-ID	Title	Description	Steps	Expected Result	Status
TC-4	Bias Spot Check	Check behavior across gender/voice variations.	1. Use balanced sample set. 2. Compare outputs by demographic variation.	No obvious systematic skew for equivalent emotional content.	Planned

2.5. Test Suite 05: Noise-Level Alerting

This suite validates the safety-oriented alert capability described in the proposal's objectives and business success criteria.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Threshold Detection	Verify alert trigger above configured dB threshold.	1. Play calibrated sounds above threshold. 2. Measure trigger time.	Alert is raised within target response time.	Planned
TC-2	No False Alarm Below Threshold	Confirm no alert for safe noise levels.	1. Play sounds below threshold. 2. Observe alert status.	No hazardous-noise alert is generated.	Planned
TC-3	Sustained Noise	Check repeated or sustained hazardous sound handling.	1. Maintain level above threshold. 2. Observe alert cadence.	User is alerted without uncontrolled spamming or lockup.	Planned
TC-4	Recovery After Noise Drop	Validate alert clearing/hysteresis logic.	1. Remove hazardous sound. 2. Observe state reset.	Alert clears appropriately and system returns to monitoring state.	Planned
TC-5	Multiple Alert Modalities	Verify vibration/LED/app notification synchronization.	1. Trigger hazardous-noise event. 2. Observe all outputs.	All configured alert channels reflect the same event state.	Planned

2.6. Test Suite 06: Mobile App Connectivity and User Interface

This suite verifies the mobile or web interface used for live captions, emotional context, settings, alerts, and user feedback.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	BLE Pairing	Verify first-time pairing with the mobile app.	1. Start device and app. 2. Pair over BLE/Wi-Fi.	Pairing completes successfully and device appears in app.	Planned
TC-2	Live Data Streaming	Check caption/emotion/alert data flow to app.	1. Run live speech session. 2. Observe app updates.	App receives and displays live events without notable desynchronization.	Planned
TC-3	Reconnect After Drop	Validate recovery from wireless interruption.	1. Force disconnect. 2. Restore link.	System reconnects or guides user clearly to reconnect.	Planned
TC-4	Settings Persistence	Verify user preferences are saved.	1. Change thresholds or UI settings. 2. Restart app/device.	Configured settings persist correctly.	Planned

TC-ID	Title	Description	Steps	Expected Result	Status
TC-5	Responsive Layout	Validate usability on supported phone sizes.	1. Open screens on target resolutions. 2. Check layout and controls.	UI remains readable, accessible, and functional.	Planned

2.7. Test Suite 07: Health Monitoring and Health Alerts (If Included in Final Build)

The proposal's later methodology and architecture sections introduce heart rate, SpO2, and temperature sensing. This suite is conditionally executed only if those modules remain in the final implementation.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Health Sensor Readings	Validate HR/SpO2/temperature acquisition if implemented.	1. Enable health module. 2. Compare readings to reference device.	Readings are captured, formatted, and transmitted correctly for prototype use.	Planned
TC-2	Abnormal Reading Alert	Verify threshold-based health alert behavior.	1. Simulate out-of-range values. 2. Observe app/device response.	System raises clear alert without app crash or data corruption.	Planned
TC-3	Missing Sensor Handling	Check graceful behavior if health sensor is absent.	1. Disable/disconnect sensor. 2. Open monitoring screen.	System reports unavailable data clearly and continues core hearing functions.	Planned

2.8. Test Suite 08: History

This suite covers local logging, optional cloud backup, and retrieval of historical captions or alerts where these features are implemented.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	History Storage	Verify that captions/events are saved when history is enabled.	1. Complete session. 2. Open history/log screen.	Stored records appear with timestamps and correct metadata.	Planned
TC-2	Cloud Sync	Validate sync behavior when network is available.	1. Generate records offline and online. 2. Trigger sync.	Data syncs correctly without duplication or corruption.	Planned
TC-3	Offline Behavior	Check local operation without internet.	1. Disable internet. 2. Run caption/alert session.	Core assistive behavior continues according to design intent.	Planned
TC-4	Data Retrieval Error Handling	Verify response to backend or database failures.	1. Simulate storage failure. 2. Access history.	System shows clear error and preserves existing data where possible.	Planned

2.9. Test Suite 09: Security, Privacy, and Input Validation

This suite addresses confidentiality, safe input handling, and prototype-appropriate privacy controls for voice, caption, and optional health data.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Secure Transport	Verify encryption of app-device/cloud communications where supported.	1. Inspect BLE/Wi-Fi/API traffic. 2. Review transport configuration.	Sensitive data is not exposed in plain text beyond intended local channels.	Planned
TC-2	Input Validation	Check handling of malformed settings, packets, or uploads.	1. Send invalid values or corrupted payloads.	System rejects invalid input gracefully with no crash.	Planned
TC-3	Unauthorized Access	Validate restricted access to settings/logs.	1. Attempt access without pairing/auth.	Unauthorized operations are blocked or clearly limited.	Planned
TC-4	Data Minimization	Verify only required user data is stored.	1. Review saved records and API fields.	Prototype stores only necessary data for functionality and testing.	Planned

2.10. Test Suite 10: Performance, Stability, and Power

This suite measures whether the prototype remains responsive, stable, and power-aware during realistic operation.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	End-to-End Latency	Measure audio-to-caption and alert latency.	1. Use timestamped test script. 2. Capture device/app timing.	Measured latency meets prototype target thresholds.	Planned
TC-2	Concurrent Event Handling	Check system behavior when captions and alerts occur together.	1. Generate speech plus threshold event. 2. Observe pipeline.	System remains stable and prioritizes critical alerts correctly.	Planned
TC-3	Extended Runtime	Validate behavior over long continuous session.	1. Run representative workload for endurance window. 2. Monitor logs.	No crash, memory leak, unsafe temperature rise, or major drift.	Planned
TC-4	Battery and Power Recovery	Check low-battery and brownout behavior.	1. Lower supply/battery state. 2. Observe warnings and recovery.	System warns appropriately and recovers cleanly after recharge/power restore.	Planned

2.11. Test Suite 11: Accessibility and Usability Validation

This suite ensures the prototype is not only technically correct but usable for the intended audience and understandable in assistive scenarios.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Caption Readability	Validate text size, contrast, and readability on app screens.	1. Review captions on target devices and lighting conditions.	Captions are readable and visually clear.	Planned
TC-2	Alert Understandability	Check whether users understand visual/vibration warnings.	1. Run guided scenarios. 2. Ask users to interpret alerts.	Users correctly interpret alerts with minimal confusion.	Planned

TC-ID	Title	Description	Steps	Expected Result	Status
TC-3	Workflow Simplicity	Assess onboarding, pairing, and routine use.	1. Give new user a short task list. 2. Observe completion.	Users can complete common tasks without heavy assistance.	Planned
TC-4	Assistive UX Feedback	Collect comfort and usefulness ratings.	1. Conduct pilot user session. 2. Record questionnaire responses.	Major UX pain points are identified and prioritized for fix.	Planned

2.12. Test Suite 12: Reliability, Recovery, and Field Conditions

This suite validates resilience under the kinds of interruptions and environmental variability expected during demonstrations or pilot evaluation.

TC-ID	Title	Description	Steps	Expected Result	Status
TC-1	Restart Recovery	Verify system recovers after reboot during active session.	1. Start session. 2. Force restart. 3. Reopen app/device.	System returns to stable state with clear recovery behavior.	Planned
TC-2	Intermittent Connectivity	Check resilience under unstable wireless conditions.	1. Vary signal quality. 2. Observe event delivery and UI state.	System degrades gracefully and avoids data corruption.	Planned
TC-3	Environmental Variation	Validate performance across quiet room, classroom, road, and office conditions.	1. Execute scenario scripts in multiple environments.	System remains functionally usable within declared prototype limits.	Planned

3. Requirement Traceability and Coverage

The QA plan is mapped to the goals and success criteria stated in the proposal so that every major feature has a corresponding validation path.

Proposal Objective / Success Area	Mapped QA Coverage	Evidence Expected
BO-01 / Affordability and functional prototype	Hardware bring-up, integration, power, and stability suites	Confirms a working low-cost demonstrator can be assembled and operated reliably
BO-02 / Sound clarity and DSP	Audio capture and noise filtering suite	Measures filtered speech quality and verifies improvement over baseline
BO-03 / Captions and emotion detection	Speech-to-text and emotion suites	Validates accuracy, latency, bounded output behavior, and UI presentation
BO-04 / Hazardous-noise alert	Noise-level alerting suite	Measures threshold detection, response time, and false-alert control

Proposal Objective / Success Area	Mapped QA Coverage	Evidence Expected
BO-05 / Open-source applicability	Configuration, logging, and traceable test evidence	Ensures reproducible builds, documented behavior, and repeatable verification steps
BO-06 / Usability testing	Accessibility and usability validation suite	Collects user feedback and confirms overall usefulness of the prototype

4. Defect Management and Reporting

All defects discovered during execution will be logged with a unique ID, build version, test environment, reproducibility notes, evidence, and owner assignment. Defects will be classified using the following severity scale.

Severity	Definition
Critical	Core assistive function unavailable, unsafe behavior, data loss, persistent crash, or false output that could mislead the user in a safety-relevant context.
High	Major feature failure, repeated disconnects, severe latency, large caption errors, or blocking UI defects with no reasonable workaround.
Medium	Feature partially works but with degraded performance, inaccurate edge behavior, recoverable errors, or noticeable usability issues.
Low	Minor cosmetic or non-blocking issues that do not materially affect safe prototype demonstration.

5. Test Environment and Data Preparation

Planned QA execution will use version-controlled firmware and app builds, at least one primary prototype device plus one backup unit, reference smartphones, calibrated or reference audio samples, noise playback tracks, standard speech scripts, and, where applicable, a comparison device for health sensors.

- Laboratory quiet-room environment for baseline capture and caption tests.
- Controlled noisy scenes such as office chatter, cafeteria noise, street ambience, and alarm-like sound samples.
- Multiple speaker profiles covering different accents, speaking speeds, and vocal intensities.
- Reference datasets for emotion clips and known caption phrases for repeatable measurement.
- Network scenarios including stable link, intermittent link, and offline mode.

6. Deliverables from QA Execution

- Approved QA Plan
- Detailed test case sheet or execution log
- Defect register with severity and status tracking
- Traceability matrix to proposal objectives

- Final QA summary report with pass/fail status, residual issues, and recommendations

7. Conclusion

This QA Plan converts the supplied proposal into a practical validation framework for an AI-assisted hearing aid prototype. It preserves the professional structure of the reference template while replacing its generic platform tests with suites that fit the actual project: audio capture, DSP, captions, emotion detection, safety alerts, connectivity, usability, privacy, and prototype reliability. Used properly, this plan will help the team demonstrate not only that AuralAid works, but that it works in a controlled, measurable, and defensible way.

Appendix A - Source Basis Used for This Plan

This document was tailored using two user-supplied files:

- Project proposal: AuralAid - Adaptive AI Hearing with Contextual Understanding.
- Reference QA template PDF: structure, section naming, and table style reference.